

Neural Network approximation of the Black-Scholes price

We consider the Black-Scholes model for a European call option with input parameters $x = (S_0, K, T, r, \sigma) \in \mathbb{R}^5$, where S_0 is the initial stock price, K is the strike, T is the time to maturity, r is the risk-free interest rate, and σ is the volatility. The Black-Scholes call price for an European option is given by the closed-form formula

$$\pi(x) = \pi(S_0, K, T, r, \sigma) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (1)$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Here Φ denotes the cumulative distribution function of a standard normal random variable. Given labeled training data $\{(x_i, \pi(x_i))\}_{i=1}^N$, we compute the empirical means $m_x \in \mathbb{R}^5$ and standard deviations $s_x \in \mathbb{R}^5$, and define

$$y = \frac{x - m_x}{s_x},$$

where the division is coordinate-wise. If m_π and s_π denotes the empirical mean and standard deviation of the training prices, let

$$\varpi(x) = \frac{\pi(x) - m_\pi}{s_\pi}.$$

We now consider a neural network with 2 layers both of width 16. We will consider the softplus activation function, defined by $\phi(u) = \log(1 + e^u)$, applied coordinate-wise. The output layer is linear, so the network defines a function

$$\widehat{\varpi}(x) = W_3 \phi \left(W_2 \phi (W_1 y + b_1) + b_2 \right) + b_3, \quad (2)$$

where $W_1 \in \mathbb{R}^{16 \times 5}$, $W_2 \in \mathbb{R}^{16 \times 16}$, $W_3 \in \mathbb{R}^{1 \times 16}$, $b_1, b_2 \in \mathbb{R}^{16}$, and $b_3 \in \mathbb{R}$ are the parameters. The parameters are chosen to minimize the mean squared error,

$$\mathcal{L}(W_1, b_1, W_2, b_2, W_3, b_3) = \frac{1}{N} \sum_{i=1}^N (\widehat{\varpi}(x_i) - \varpi(x_i))^2. \quad (3)$$

Then we get the original price as

$$\widehat{\pi}(x) = m_\pi + s_\pi \widehat{\varpi}(x). \quad (4)$$

To test the network and explain the predictions, we use SHAP values, which decompose the prediction into contributions from each input feature. In rough strokes, the SHAP values $\varphi_j(x)$ for $j = 1, \dots, 5$ are defined such that

$$\widehat{\pi}(x) \approx \mathbb{E}[\widehat{\pi}(X)] + \varphi_{S_0}(x) + \varphi_K(x) + \varphi_T(x) + \varphi_r(x) + \varphi_\sigma(x),$$

where $\varphi_j(x)$ measures the contribution of feature j to the predicted price relative to a background distribution of inputs. E.g., if $\varphi_\sigma(x)$ is positive, then the volatility input increases the predicted call price relative to the background average. For a well-trained Black-Scholes neural network, the SHAP values should reflect the financial intuition that call prices increase with S_0 and σ , and decrease with K . With r and T the financial intuition is more subtle.